

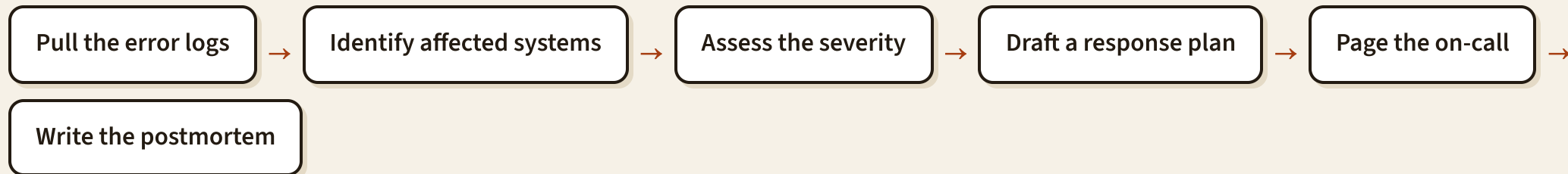
Architecture Over Intelligence

Designing reliable AI workflows

Amar Patel · The AI Fellowship Madrid · 9 Oct 2026



"Investigate this incident" is one job. *It isn't.*



Six steps. Each one feeds the next.

The demo vs. production gap

Demo

- One uninterrupted session
- Mistakes are annoying
- A human watches the whole time
- A frontier model

Production

- Compaction, reset, walk-aways
- Mistakes are data loss
- A human checks hours later
- A cheaper model that drifts

Reliability fails at the boundaries.

A JSON file is the *source of truth*

● process-state.json		meeting_id: 2026-01-15_140000
current_step	2.6 → next is 3	
completed_steps	[-1, 0, 1, 2, 2.5, 2.6]	
user_confirmed.summary	true	
aborted	false	

Before every step, read the file. A step runs only when `current_step` is the one before it.

A gate is *authority*, not just quality control



Auto

Safe, predictable steps keep moving.
File reads, health checks.



Human

The agent stops and waits.
Anything that needs a yes.



Blocked

Cannot proceed.
Missing file, corrupted state.

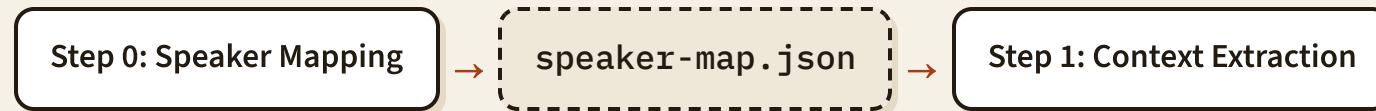


REVIEWS EVERY STEP

GUARDS ONLY THE IRREVERSIBLE

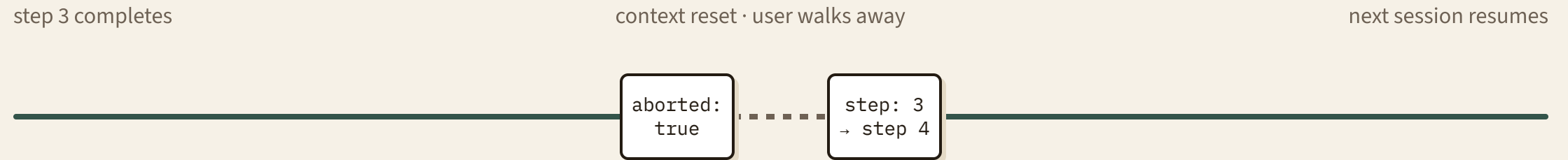
The human is a **dial, not a switch.**

No step uses data that doesn't exist yet



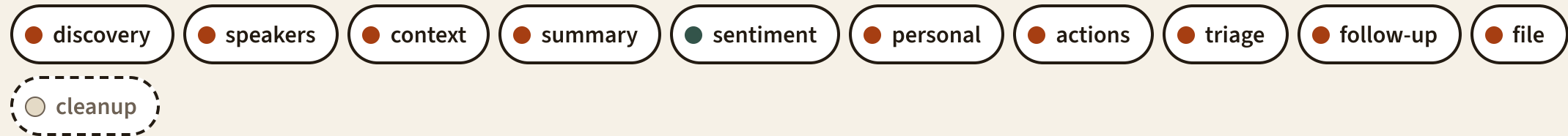
Declare what you produce. Gate on what you consume.

Interruption is not a bug



Resume from the same file. It survives compaction, a context reset, and your absence.

One real pipeline: *11 steps, 8 LLM calls*



9

HUMAN GATES

4

SCRIPTS

1

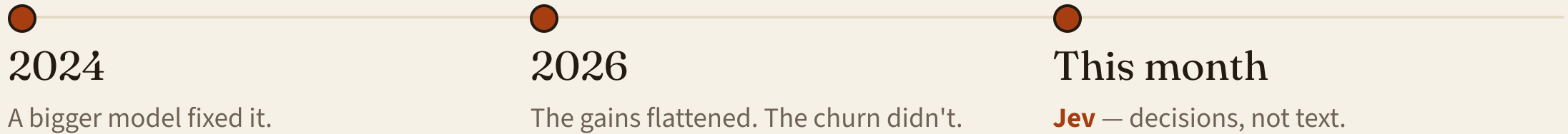
STATE FILE

What broke in three months

- ✎ The state file got edited to the wrong step → *validate on load*
- ⏮ The LLM skipped a gate → *the file decides, not the prompt*
- ✓ Compaction mid-pipeline → *the file survived* — the win

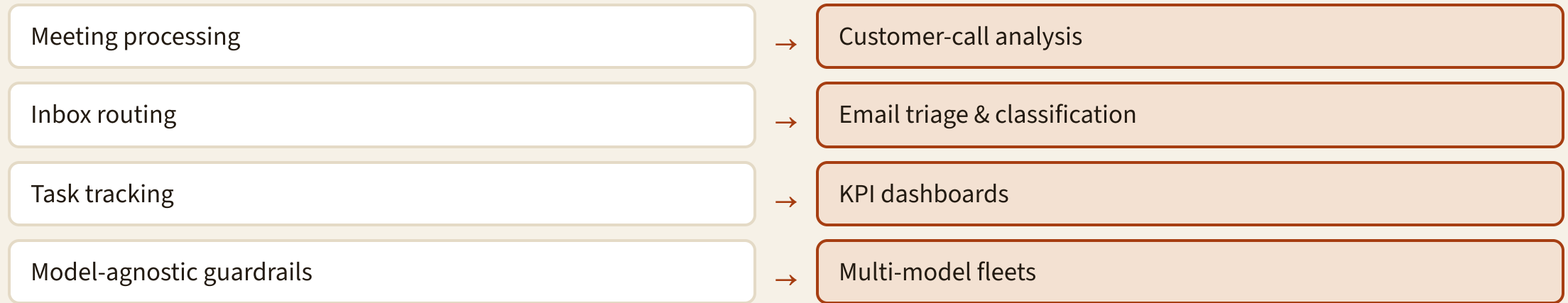
Never assume a step is done.

The model is always changing



“The State Machine Is the Agent.”

These patterns work everywhere



The domain changes. The pattern doesn't.

Five rules

- 1 The **state file** is the source of truth.
- 2 **Gates** are structural, not optional.
- 3 Declare **artifacts** before they're needed.
- 4 Design for **interruption**.
- 5 **Architecture > intelligence.**

Key takeaways

- The model is a variable. The structure is the constant.
- The human is a **dial, not a switch** — better models move it up the stakes curve.
- It runs on any model. That part comes free.



REVIEWS EVERY STEP

GUARDS ONLY THE IRREVERSIBLE

Reliability at the boundaries.